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(54) **MANAGEMENT OF COMMUNICATIONS FOR OVERLAPPING SUBNETS USING IPV6 ADDRESSING**

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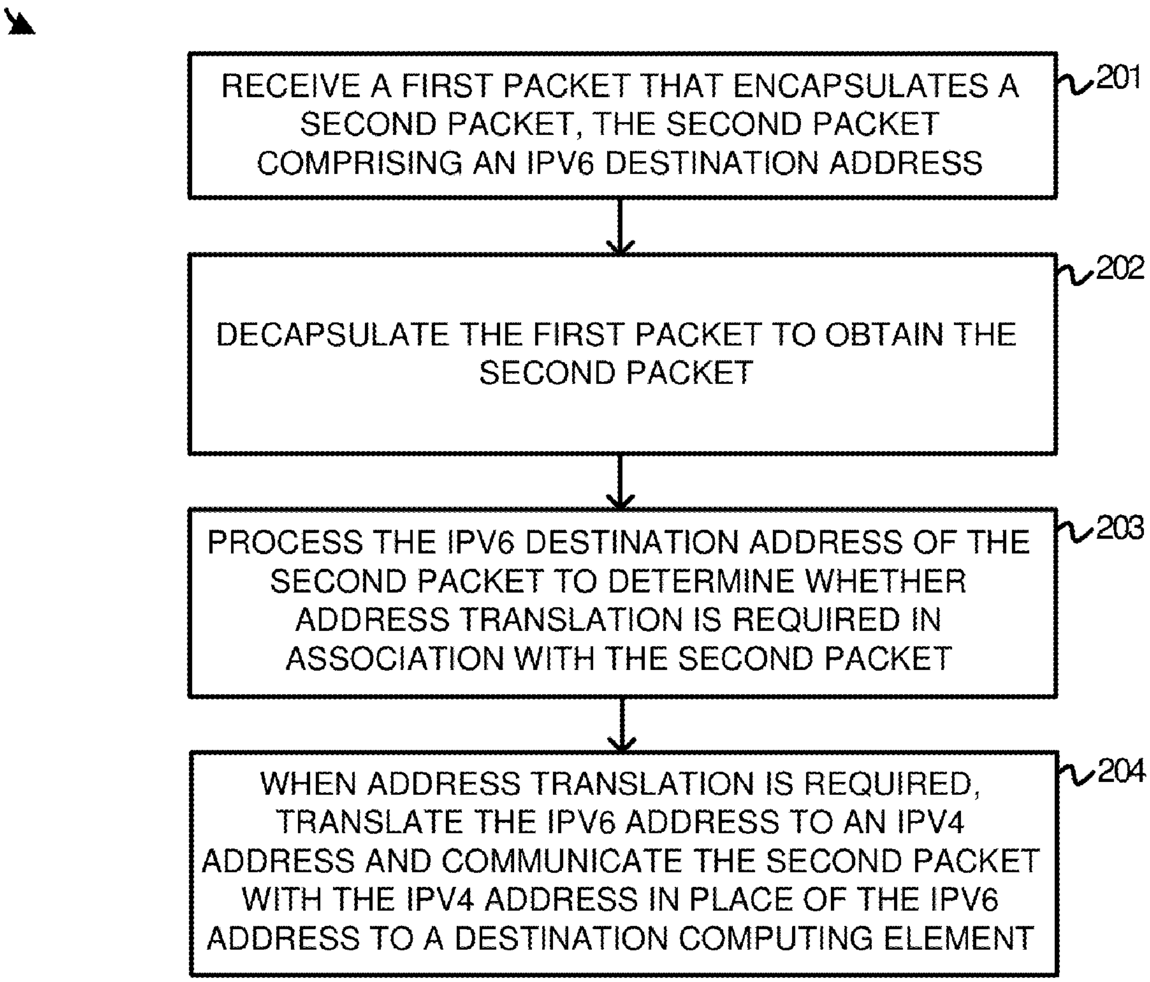
(57) **ABSTRACT**

This technology manages overlapping subnets through IPv6-to-IPv4 translation. A method involves an application on a source device obtaining an IPv6 destination address from a coordination service. This address includes an IPv4 destination and a gateway identifier. The application generates a packet to this IPv6 address and sends it to a private networking service on the source device. The service uses the gateway identifier to determine the gateway's public IP address, as indicated by the coordination service. It then encapsulates the packet and sends it to the gateway at the public IP address.

200



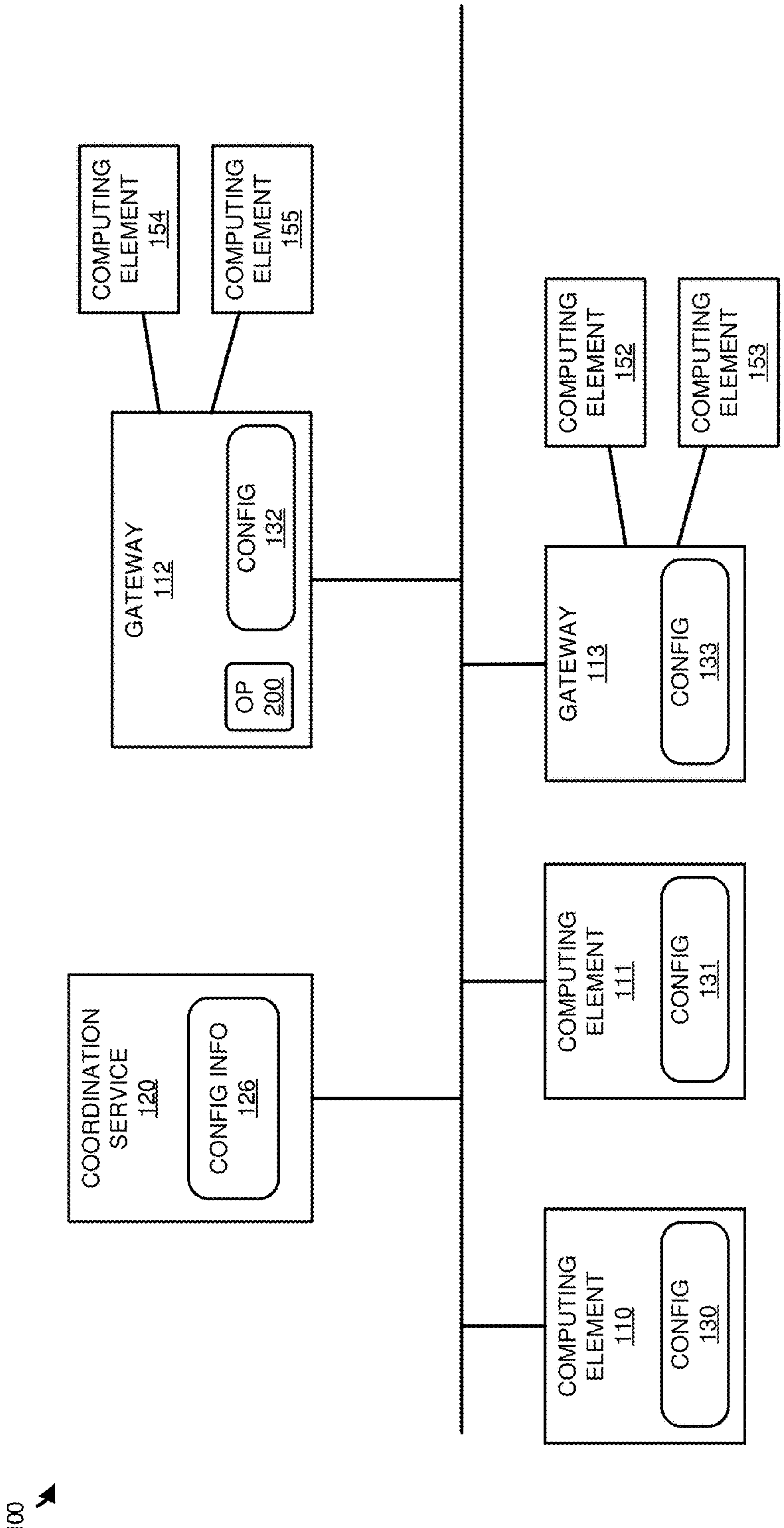

100
COORDINATION SERVICE 120
CONFIG INFO 126
GATEWAY 112
OP 200
CONFIG 132
COMPUTING ELEMENT 154
COMPUTING ELEMENT 155
COMPUTING ELEMENT 110
CONFIG 130
COMPUTING ELEMENT 111
CONFIG 131
GATEWAY 113
CONFIG 133
COMPUTING ELEMENT 152
COMPUTING ELEMENT 153

FIGURE 1

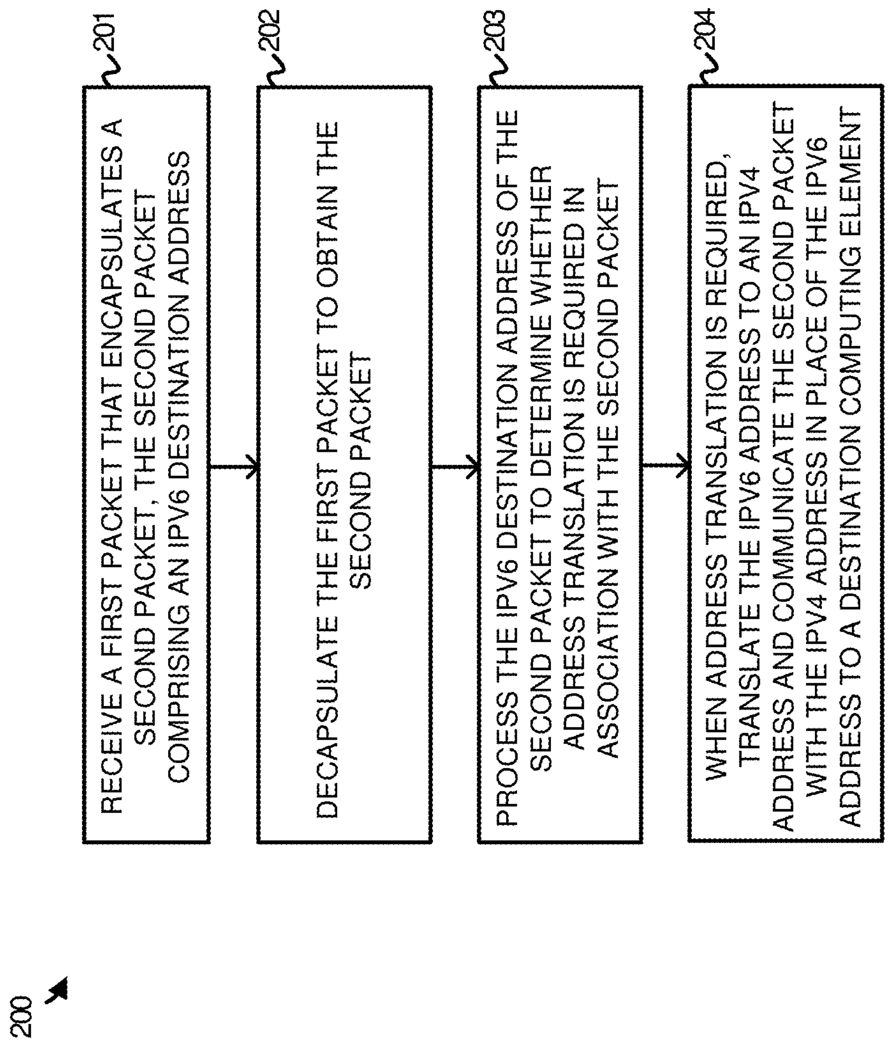

200
201
RECEIVE A FIRST PACKET THAT ENCAPSULATES A SECOND PACKET, THE SECOND PACKET COMPRISING AN IPV6 DESTINATION ADDRESS
202
DECAPSULATE THE FIRST PACKET TO OBTAIN THE SECOND PACKET
203
PROCESS THE IPV6 DESTINATION ADDRESS OF THE SECOND PACKET TO DETERMINE WHETHER ADDRESS TRANSLATION IS REQUIRED IN ASSOCIATION WITH THE SECOND PACKET
204
WHEN ADDRESS TRANSLATION IS REQUIRED, TRANSLATE THE IPV6 ADDRESS TO AN IPV4 ADDRESS AND COMMUNICATE THE SECOND PACKET WITH THE IPV4 ADDRESS IN PLACE OF THE IPV6 ADDRESS TO A DESTINATION COMPUTING ELEMENT

FIGURE 2

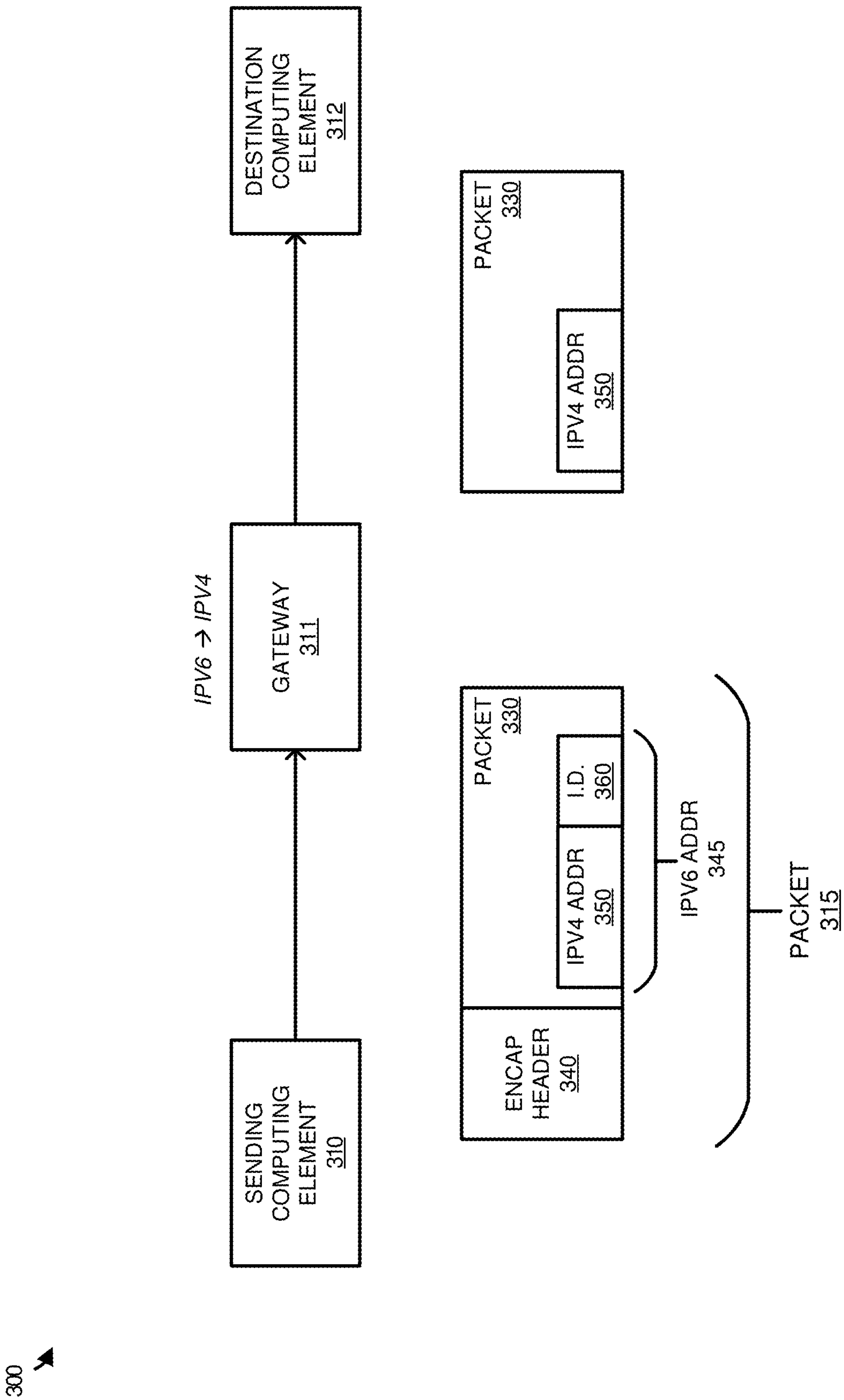

300
SENDING COMPUTING ELEMENT 310
IPV6 → IPV4
GATEWAY 311
DESTINATION COMPUTING ELEMENT 312
ENCAP HEADER 340
PACKET 330
IPV4 ADDR 350
I.D. 360
IPV6 ADDR 345
PACKET 315
PACKET 330
IPV4 ADDR 350

FIGURE 3

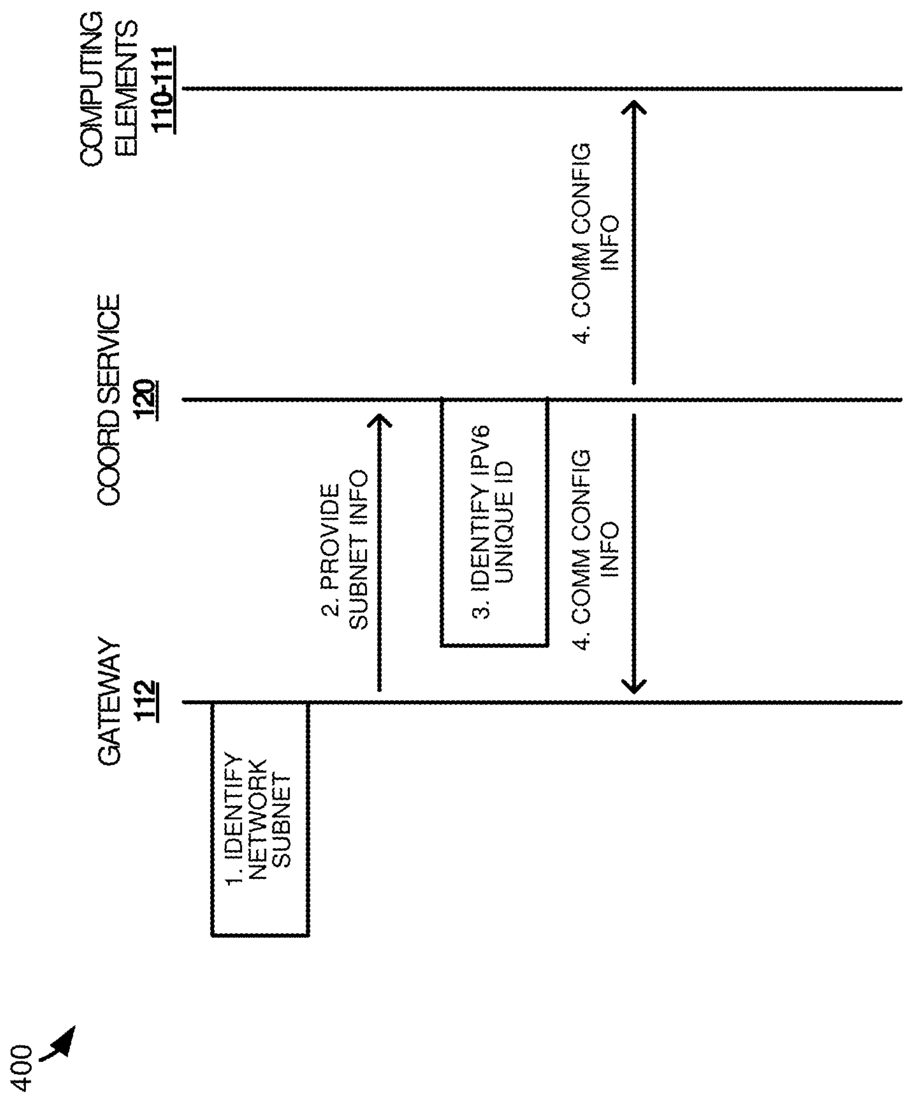

400
GATEWAY
112
COORD SERVICE
120
COMPUTING
ELEMENTS
110-111
1. IDENTIFY
NETWORK
SUBNET
2. PROVIDE
SUBNET INFO
3. IDENTIFY IPV6
UNIQUE ID
4. COMM CONFIG
INFO
4. COMM CONFIG
INFO

FIGURE 4

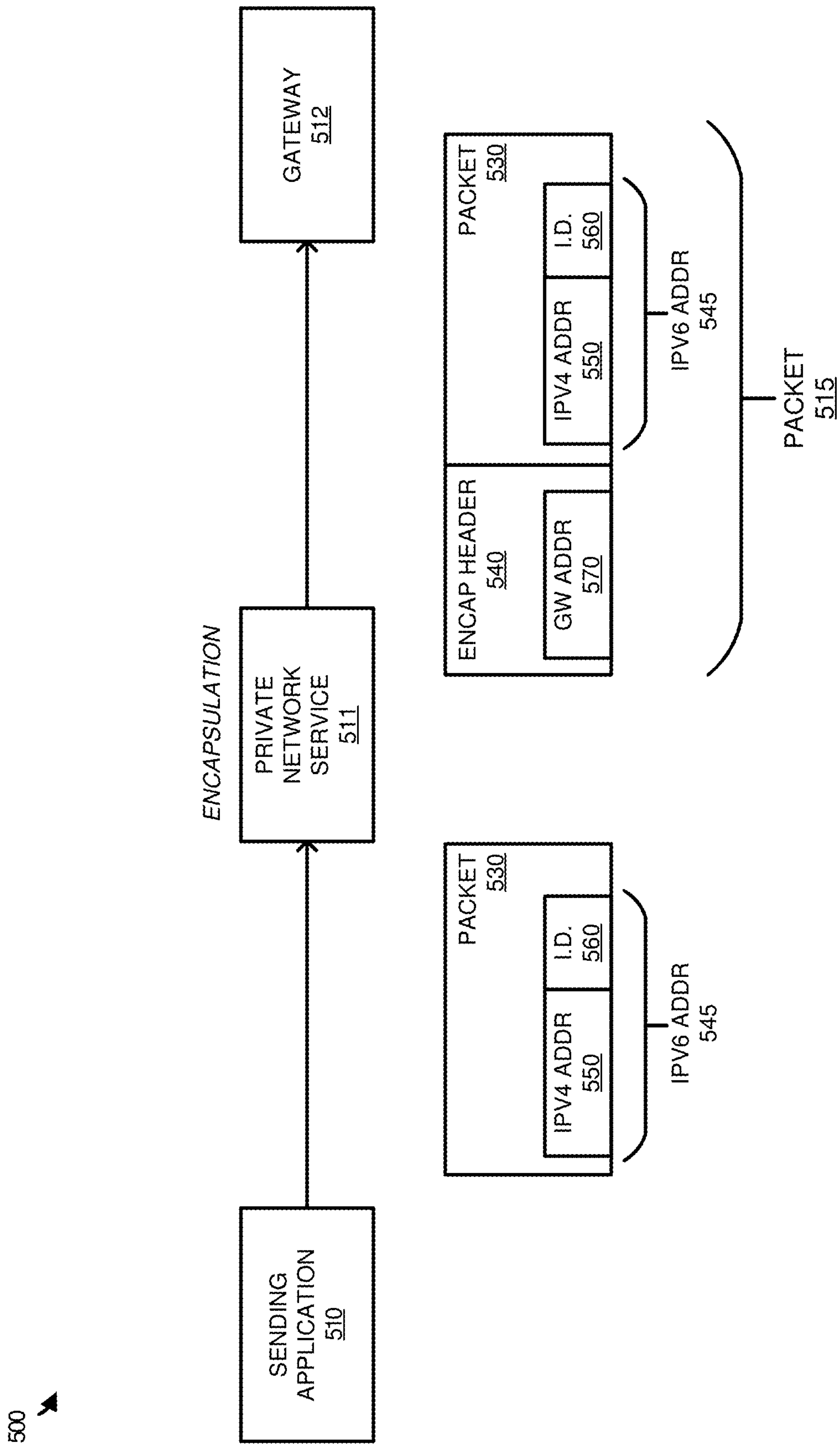

500
SENDING APPLICATION 510
ENCAPSULATION
PRIVATE NETWORK SERVICE 511
GATEWAY 512
PACKET 530
IPV4 ADDR 550
I.D. 560
IPV6 ADDR 545
ENCAP HEADER 540
GW ADDR 570
PACKET 530
IPV4 ADDR 550
I.D. 560
IPV6 ADDR 545
PACKET 515

FIGURE 5

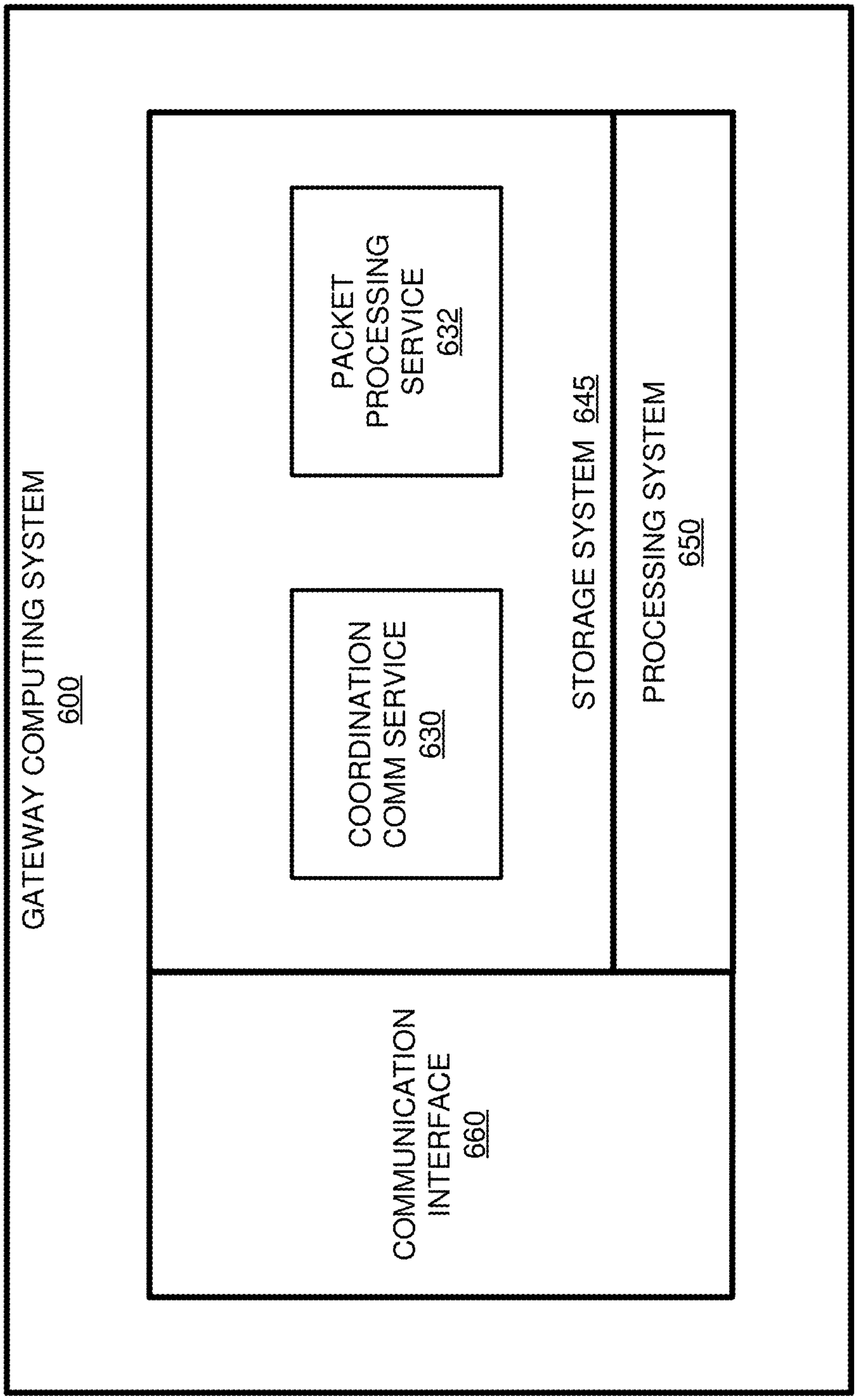

GATEWAY COMPUTING SYSTEM
600
COORDINATION
COMM SERVICE
630
PACKET
PROCESSING
SERVICE
632
STORAGE SYSTEM 645
PROCESSING SYSTEM
650
COMMUNICATION
INTERFACE
660

FIGURE 6

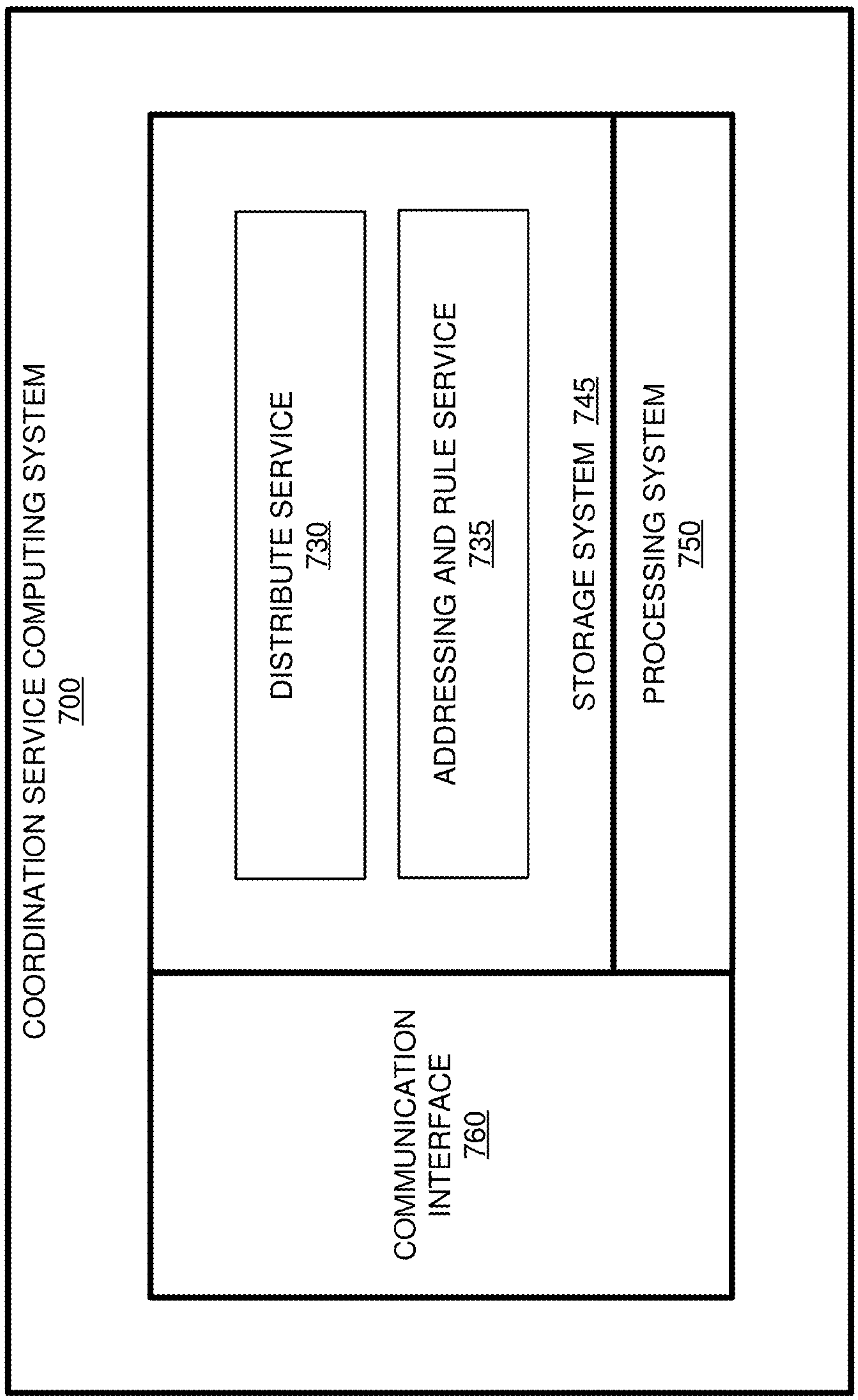

COORDINATION SERVICE COMPUTING SYSTEM
700
DISTRIBUTE SERVICE
730
ADDRESSING AND RULE SERVICE
735
STORAGE SYSTEM 745
PROCESSING SYSTEM
750
COMMUNICATION
INTERFACE
760

FIGURE 7

MANAGEMENT OF COMMUNICATIONS FOR OVERLAPPING SUBNETS USING IPV6 ADDRESSING

RELATED APPLICATIONS

[0001] This application is a continuation of and claims priority to U.S. Pat. No. 12,309,114, entitled "MANAGEMENT OF COMMUNICATIONS FOR OVERLAPPING SUBNETS USING IPv6 ADDRESSING," filed on Jan. 20, 2023, which is related to and claims priority to U.S. Provisional Patent Application No. 63/391,229, entitled "MANAGEMENT OF COMMUNICATIONS FOR OVERLAPPING SUBNETS USING IPv6 ADDRESSING," filed on Jul. 21, 2022, and which are both hereby incorporated by reference in their entirety.

TECHNICAL BACKGROUND

[0002] In computing networks, physical and virtual computing systems can include applications and services that require communications with other computing systems to provide desired operations. For example, an application on a first computing system may require data from a storage server located on a second computing system. To provide the communication, the data payload may be placed in a network packet and transferred to the required computing system. However, although network packets provide a method of communication between computing systems, difficulties often arise in maintaining security and configuration information to support the communications.

[0003] To overcome some of the deficiencies presented in securing network communications, various technologies have been developed. These technologies include virtual local area networks (VLANs), encryption for the data payload within the data packets, amongst other similar security procedures. Yet, while these security technologies may provide additional security over unprotected network packets, configuring private networks can be difficult and cumbersome. These difficulties are compounded as the number of virtual and physical computing elements are increased in a network, requiring increased management in association with addressing for the computing elements.

SUMMARY

[0004] The technology described herein manages overlapping subnets using IPv6 to IPv4 translation. In one example, a method includes, in an application executing on a source computing element, obtaining an IPv6 destination address for a destination on the private network from a coordination service of the private network. The IPv6 destination address includes an IPv4 destination address of the destination in an IPv4 subnet and a gateway identifier identifying a gateway to the IPv4 subnet. The method further includes, in the application, generating the packet addressed to the IPv6 destination address and forwarding the packet to a private networking service on the source computing element. The method also includes, in the private networking service, determining a public IP address for the gateway from the gateway identifier. The coordination service indicates the public IP address corresponding to the gateway identifier. Also, in the private networking service, the method includes encapsulating the packet to create an encapsulated packet addressed to the public IP address and sending the encapsulated packet to the gateway at the public IP address.

[0005] An apparatus and one or more computer readable storage media are provided in other examples to perform steps similar to those described in the method above.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** illustrates a computing environment to manage overlapping subnets in a private network using IPv6 to IPv4 translation according to an implementation.

[0007] FIG. **2** illustrates an operation of a gateway to manage overlapping subnets in a private network using IPv6 to IPv4 translation according to an implementation.

[0008] FIG. **3** illustrates an overview of packet flow for a gateway to provide IPv6 to IPv4 translation according to an implementation.

[0009] FIG. **4** illustrates a timing diagram of requesting and distributing configuration information for a private network according to an implementation.

[0010] FIG. **5** illustrates an overview of packet generation in a private network according to an implementation.

[0011] FIG. **6** illustrates a gateway computing system to manage overlapping subnets in a private network using IPv6 to IPv4 translation according to an implementation.

[0012] FIG. **7** illustrates a coordination service computing system to manage and distribute configuration information associated with a private network according to an implementation.

DETAILED DESCRIPTION

[0013] FIG. **1** illustrates a computing environment **100** to manage overlapping subnets in a private network using IPv6 to IPv4 translation according to an implementation. Computing environment **100** includes coordination service **120**, computing elements **110-112** and **152-155**, and gateways **112-113**. Coordination service **120** may use one or more computers to support the management and distribution configuration information **126**. Coordination service **120** can be implemented as a virtual machine on a computer in some examples. Computing elements **110-111** and gateways **112-113** include configurations **130-133**. Gateway **112** is further configured to perform operation **200** that is described below in FIG. **2**. Gateways **111-112** can be representative of routers capable of connecting computing elements **152-155** with other computing elements via the internet.

[0014] In computing environment **100**, coordination service **120** supports a private network using configuration information **126**. When a computing element requests to join the private network, the computing element can communicate a request with credentials to coordination service **120**, wherein the credentials can include identifier information associated with the requesting computing element. In response to the request, coordination service **120** may verify the requesting computing element and provide communication configuration to the computing element, permitting the computing element to communicate with one or more other members of the private network. For example, computing element **110** can represent a computer that attempts to join a private corporate network. In response to the request from computing element **110**, coordination service **120** can determine whether computing element **110** is permitted to join the network and provide configuration **130** to computing element **110** when computing element is permitted to join the network. Configuration **130** can include encryption keys to communicate with other computing elements (and gateways), public IP addressing information, private IP addressing information, or some other information to support the communications in the network. In some examples, the information that is provided is based on the identity of the computing element and/or user of the computing element that is supplied as part of the credentials.

[0015] When a communication is requested by computing element **110** to computing element **111**, the communication can use a private destination IP address associated with computing element **111** and private source IP address associated with computing element **110**. The destination IP address can be obtained using a DNS request, wherein a domain can be translated to the requisite destination IP address using configuration **130**. Once the communication with the private IP addresses is generated, the communication is encapsulated using an encryption key associated with computing element **111** and communicated over the network using the public IP addressing information associated with computing elements **110-111**.

[0016] In some implementations, computing elements, such as computing elements **152-155**, can use a gateway to support the communications over the private network. Rather than decapsulating or encapsulating packets at the individual computing element, the gateway can support the encapsulation and decapsulation of packets. For example, when an encapsulated packet is received at gateway **112**, gateway **112** can decapsulate the packet and forward the packet to a destination computing element of computing elements **154-155**. Computing elements **154-155** can receive the packet without identifying the existence of the private network.

[0017] In the examples of gateways **112-113**, gateways **112-113** may communicate with coordination service **120** to indicate subnets that are available behind the gateway. For example, gateway **112** can support virtual machines, containers, and physical devices that use an IPv4 subnet. Gateway **113** can also indicate an IPv4 subnet that is available behind the gateway. When the subnets overlap, coordination service **120** can assign a unique identifier to each of gateways **112-113** that directs packets to the correct gateway. Specifically, rather than using the IPv4 address associated with a computing element, the packets can use an IPv6 address that indicates the IPv4 address for the computing element and the unique identifier for gateway **112**. Accordingly, if computing element **110** generated a packet for delivery to computing element **154**, the packet can use the IPv6 address associated with computing element **154**. The packet is then encapsulated at computing element **110** and communicated to a public IP address for gateway **112**. The encapsulation and use of the public IP address for gateway **112** can be defined in configuration **130**. Once the packet is received at gateway **112**, the packet is decapsulated, the IPv6 address is translated to an IPv4 address, and the packet is forwarded to computing element **154**. Advantageously, while multiple gateways (which can each represent virtual private clouds) can support computing elements that use the same subnet, IPv6 addressing can be used to define the appropriate destination gateway for a packet. In some examples, in translating the IPv6 address to the IPv4 address, gateway **112** can convert the IPv6 header for the packet to an IPv4 header. The IPv4 header can comprise at least the destination IPv4 address and can further include a source IPv4 address in some examples.

[0018] In some implementations, the sending computing element **110** can manually enter the IPv6 address, however, sending computing element may use a domain name system (DNS) associated with the private network to translate a domain into the IPv6 address. The DNS can be local to computing element **110** and provided by coordination service **120** or can be located on another system available to computing element **110**. When a packet is generated, the packet is identified by a service on computing element **110**, wherein the service can identify the IPv6 address and encapsulate the packet based on the IPv6 address. The encapsulation may include encrypting using a key associated with gateway **112**, adding public addressing information for gateway **112** and computing element **110**, or some other operation. Once encapsulated, the encapsulated packet can be forwarded to gateway **112**.

[0019] FIG. **2** illustrates an operation **200** of a gateway to manage overlapping subnets in a private network using IPv6 to IPv4 translation according to an implementation. The steps of operation **200** are referenced parenthetically in the paragraphs that follow with reference to systems and elements of computing environment **100**. Although demonstrated using gateway **112**, similar operations can be performed by gateway **113** to support communications with computing elements **152-153**.

[0020] In operation **200**, gateway **112** receives (**201**) a first packet that encapsulates a second packet, the second packet comprising an IPv6 address. In some implementations, computing elements in a private network are assigned private IP addresses to support communications between the devices. For virtual private clouds (VPCs), computing elements behind the private cloud can be associated with a subnet. For example, computing elements **154-155** can correspond to a first subnet and computing elements **152-153** can correspond to a second subnet, wherein at least a portion of computing elements **152-155** can share the same IPv4 address. To prevent communications being directed at the wrong computing node, gateways **112-113** can each be assigned a unique identifier by coordination service **120**. The unique identifier can be included as part of an IPv6 address that also includes the IPv4. Thus, the VPC can be uniquely identified along with the IPv4 subnet address for the destination computing element. For example, a packet communicated by computing element **110** directed at computing element **154** can use the IPv6 address associated with computing element **154** as the destination IP address. Once the packet is generated, computing element **110** can encapsulate the packet based on configuration **130**, wherein the encapsulation can encrypt the packet using a key associated with gateway **112** and a destination public IP address associated with gateway **112** in the encapsulation header. The encapsulated packet is then forwarded to gateway **112**.

[0021] In response to receiving the first packet, operation **200** further decapsulates (**202**) the first packet to obtain the second packet and processes (**203**) the IPv6 destination address of the second packet to determine whether address translation is required in association with the second packet. Here, the decapsulation may use a private encryption key associated with gateway **112**. Once decapsulated, gateway **112** can determine whether the destination IPv6 address includes a flag or other identifier indicating the translation requirement. In other implementations, gateway **112** can be configured to translate all incoming packets from an IPv6 to an IPv4 address.

[0022] When address translation is required, operation **200** further translates (**204**) the IPv6 address to an IPv4 address and communicates the second packet with the IPv4 address in place of the IPv6 address to a destination computing element. In some examples, gateway **112** can extract the IPv4 address from the IPv6 address and forward the packet to the computing element associated with the IPv4 address. Returning to the example of computing element **110** generating a packet destined for computing element **154**, gateway **112** can replace the destination IPv6 address with the corresponding IPv4 address, wherein the address may be included in the IPv6 address. Once translated, the packet is forwarded to computing element **154**. Computing element **110** can distinguish between forwarding the packet to gateway **112** or gateway **113** based on the IPv6 address that includes the unique identifier for either gateway **112** or gateway **113**. In some examples, in replacing the IPv6 address with the IPv4 address, gateway **112** can generate a replacement IPv4 header for the packet. The IPv4 header can include at least the destination IPv4 address and can further include a source IPv4 address or some other IPv4 information.

[0023] FIG. **3** illustrates an overview **300** of packet flow for a gateway to provide IPv6 to IPv4 translation according to an implementation. Overview **300** includes sending computing element **310**, gateway **311**, and destination computing element **312**. Operational scenario **300** further includes packet **315** with encapsulation header **340** and packet **330**. Packet **330** includes IPv6 address **345** with IPv4 address **350** and identifier **360**.

[0024] In overview **500**, sending computing element **310** communicates packet **315** to gateway **311**. In generating packet **315**, an application on sending computing element **310** can identify a destination IPv6 address associated with the destination computing element. The IPv6 address can be manually entered, can be determined via a domain name system (DNS), or can be identified in some other means. In at least one example, sending computing element **310** can receive a DNS configuration that permits DNS requests to be supplied with addresses associated with other computing elements in the private network. For example, a DNS request by an application on sending computing element **310** can return IPv6 address **345**, permitting packet **330** to be generated for the application. Once packet **330** is generated, the packet can be forwarded to a private networking service that can operate as part of the operating system for sending computing element **310** or can act as a standalone process. The private networking service can identify IPv6 address **345** and encapsulate packet **330** as packet **315**. The encapsulation can use an encryption key associated with gateway **311**, can include public addressing information associated with sending computing element **310** and gateway **311**, or can include some other information. The public addressing information can include public IP addressing, public ports, or some other public addressing information. In some examples, the private networking service can determine the public addressing information and encryption information for the packet based on a configuration supplied by the coordination service. The configuration information can associate the IPv6 address **345** with public addressing information and encryption information associated with the IPv6 address. In some examples, the configuration information can associate a unique identifier **360** for gateway **311** included in the IPv6 address to select encryption and public addressing for packet **315**. Specifically, the private networking service can identify packet **330**, identifies unique identifier **360** in packet **330**, and select parameters for encapsulation based on the identifier. Advantageously, rather than maintaining a unique entry for each computing element behind gateway **311**, an address with the unique identifier for gateway **311** may be encapsulated and forwarded to gateway **311** using encryption and public addressing associated with the gateway. Thus, multiple computing elements in an IP subnet at a gateway can be encapsulated in a similar manner. Further, while demonstrated with a single gateway, other unique identifiers can be associated with other gateways in the private network. Thus, while multiple gateways can support the same private addressing subnet, outside computing elements, such as developer or administrator computing elements, can communicate using a unique IPv6 address associated with the element of interest.

[0025] After packet **315** is generated, the packet is communicated to gateway **311**. Gateway **311** may represent a router for a virtual private cloud that supports network connectivity for virtual machines, containers, and the like in the private network cloud. In response to receiving packet **315**, gateway **311** decapsulates the packet, wherein the decapsulation may use a private encryption key for gateway **311**. Once decapsulated, gateway **311** can determine whether address translation is required in association with IPv6 address **345**. In some examples, IPv6 address **345** includes a flag indicating the requirement of translation. Once identified, gateway **311** can identify IPv4 address **350** in IPv6 address **345** and replace IPv6 address **345** with IPv4 address **350** in packet **350**. Once replaced, the packet can be forwarded to the destination computing element (e.g., a virtual machine). The replacement of the IPv6 address with the IPv4 address can include replacing the IPv6 header of the packet with an IPv4 header that comprises at least the IPv4 destination address.

[0026] FIG. **4** illustrates a timing diagram **400** of requesting and distributing configuration information for a private network according to an implementation. Timing diagram **400** includes gateways **112**, coordination service **120**, and computing elements **110-111** from computing environment **100** of FIG. **1**.

[0027] In timing diagram **400**, gateway **112** identifies a network subnet in association with computing elements connected to gateway **112** at step **1**. The computing elements may comprise virtual machines, containers, physical computers, or some other computing element. In some implementations, gateway **112** can represent a router for a VPC, wherein gateway **112** connects a subnet of private addresses to a larger network. Gateway **112** then provides the subnet information to coordination service **120** at step **2**, wherein the coordination service is responsible for distributing communication configuration information to computing elements of a private network.

[0028] Once provided, coordination service **120** can identify an IPv6 unique identifier for the gateway at step **3** and distribute communication configuration information in the network to computing elements and gateways in the private network to support communication at step **4**. The configuration information can include public encryption keys, public addressing information, private addressing information, or some other information. The private addressing information can include IP addresses allocated the subnet for gateway **120**, IP addresses allocated for other computing elements in the network, available types of communications (protocols, ports, and the like), or some other information. For example, coordination service **120** can provide a unique identifier for packets communicated to gateway **112** to computing elements **110-111**, wherein computing elements **110-111** can combine the unique identifier for gateway **112** with an IPv4 address for a computing element behind gateway **112** to generate an IPv6 address for the computing element. The IPv6 address can further include an indicator that requires a translation to IPv4 address or some other information to deliver a packet to a computing element behind gateway **112**. Further, the information provided to gateway **112** may include private and public addressing information to communicate with computing elements **110-111** or some other information.

[0029] Once communication configuration information is distributed to computing elements **110-111** and gateway **112**, an application at computing element **110** can generate a packet that addresses a computing element behind gateway **112**. The packet can use an IPv6 address allocated by coordination service **120**, wherein the IPv6 address includes the IPv4 address for the subnet behind gateway **112**, a unique identifier for gateway **112**, an indication that the packet comprises a private network packet, or some other information. After generating the packet, computing element **112** can identify that the packet is a private network packet, encapsulate the packet using an encryption key and public addressing information associated with gateway **112**. Once encapsulated, the packet can be communicated to gateway **112**. In response to receiving the packet, gateway **112** can decapsulate the packet, translate the IPv6 address to the IPv4 address, and forward the packet to the destination computing element (e.g., virtual machine). In translating the address, gateway **112** can replace the IPv6 header with an IPv4 header that includes at least the destination IPv4 address to direct the packet to the destination computing element.

[0030] FIG. **5** illustrates an overview **500** of packet generation in a private network according to an implementation. Overview **500** includes sending application **510**, private network service **511**, and gateway **512**. Overview **500** further includes packet **515**, packet **530**, encapsulation header **540**, gateway addressing **570**, IPv4 addressing **550**, and identifier **560** representative of a gateway identifier for gateway **512**. Sending application **510** and private network service **511** can operate on the same computing system in some examples, wherein private network service **511** can represent the operating system or a separate service capable of providing the encapsulation operations described herein.

[0031] In overview **500**, sending application **510** generates packet **530** with IPv6 addressing **545**, wherein IPv6 addressing **545** includes IPv4 addressing **550** and identifier **560**. In some implementations, the IPv6 address can be manually entered for sending application **510**, wherein a user may identify the IPv6 address and enter the address as the destination for the packet. In other implementations, sending application **510** can generate a DNS request to a DNS associated with the private network. For example, sending application **510** may generate a DNS request with the domain associated with IPv6 address **545**. The request can be forwarded to a DNS on the same computing system as sending application **510** or can be forwarded to an external DNS. The DNS can then return IPv6 address **545** that corresponds to the unique address for the destination computing element for the packet.

[0032] Once packet **530** is generated, packet **530** is identified by private network service **511**. Private network service **511** can correspond to the operating system or another service executing on the computing system with sending application. Private network service **511** can execute on a separate system (e.g., router) in some examples. In response to identifying packet **530**, private network service **511** performs encapsulation on packet **530** based at least on the IPv6 address **545**. In some examples, private network service **511** may process the addressing attributes in packet **530** to determine whether the communication is permitted. Specifically, the addressing information can be compared to the permitted communications provided from the coordination service. Once the communication is permitted, the packet is encapsulated, wherein packet **530** is encrypted using a public encryption key associated with gateway **512** and encapsulation header **540** is added for the encapsulation header. Encapsulation header **540** includes public addressing information, including gateway address **570** that corresponds to gateway **512**. Other information in encapsulation header **540** can include public source IP addressing, public source and destination port information, protocol information, or some other information.

[0033] In some examples, in identifying the destination gateway **512** for the packet, IPv6 address **545** can indicate a unique identifier for gateway **512**. The unique identifier can be used to distinguish gateway **512** from other gateways with possible overlapping subnets. For example, gateway **512** may act as a router for a plurality of computing elements (e.g., virtual machines) that correspond to a first subnet, and a second gateway that acts as a router for a second plurality of computing elements that correspond to a second subnet, wherein the first subnet overlaps at least partially with the second subnet. To permit direct communications to the first subnet and the second subnet, the coordination service can assign or indicate a unique identifier for the gateway associated with the computing element. When the packet is generated, the IPv6 address can include a notification that translation is required from the IPv6 address to the IPv4 address when the packet is received at the destination gateway **512**. IPv6 address **545** further includes an identifier for gateway **512** and IPv4 address **550** that corresponds to IPv4 address of the destination computing element in the subnet. Accordingly, while sending application **510** can use an IPv6 address to communicate the packet to appropriate destination gateway and subnet, translation can be performed on the address at the gateway to provide the expected IPv4 address in the header of packet **530**.

[0034] After packet **515** is generated, the packet is forwarded to gateway **512**, wherein gateway **512** can represent a router for a subnet of end computing elements. Gateway **512** can decapsulate the packet by applying the private encryption key associated with gateway **512** and can process the packet to determine whether address translation is required in association with packet **530**. In some implementations, IPv6 address **545** may include a notification or flag that network address translation is required. Once the flag is identified by gateway **512**, IPv4 address **550** can be extracted from the IPv6 address **545** and used to replace IPv6 address **545** prior to forwarding to a destination computing element. In replacing the IPv6 destination address with the IPv4 destination address, the IPv6 header can be replaced with an IPv4 header that includes at least the destination IPv4 address to direct communications to the destination computing element.

[0035] In some implementations, by using unique gateway identifiers, such as identifier **560** for gateway **512**, packet processing operations can be expedited for gateway **512**. Specifically, gateway **512** identifies a flag (comprising a set of one or more bits in the IPv6 header), identifies the IPv4 address in the IPv6 address, and applies the IPv4 address in place of the IPv6 address. Accordingly, rather than translating the IPv6 address using a data structure or table, the IPv4 address can be extracted from the included IPv6 address. Additionally, the sending computing element (e.g., administrator computer) can identify the gateway identifier in the IPv6 address and identify the required public addressing information based on the gateway identifier. The public addressing information can include a public IP address associated with the gateway, a source IP address associated with the sending computing element, or some other communication information.

[0036] FIG. **6** illustrates a gateway computing system **600** to manage overlapping subnets in a private network using IPv6 to IPv4 translation according to an implementation. Computing system **600** is representative of any computing system or systems with which the various operational architectures, processes, scenarios, and sequences disclosed herein for a computing element can be implemented. Computing system **600** is an example of gateways **112-113** of FIG. **1**, although other examples may exist. Computing system **600** includes storage system **645**, processing system **650**, and communication interface **660**. Processing system **650** is operatively linked to communication interface **660** and storage system **645**. Communication interface **660** may be communicatively linked to storage system **645** in some implementations. Computing system **600** may further include other components such as a battery and enclosure that are not shown for clarity.

[0037] Communication interface **660** comprises components that communicate over communication links, such as network cards, ports, radio frequency (RF), processing circuitry and software, or some other communication devices. Communication interface **660** may be configured to communicate over metallic, wireless, or optical links. Communication interface **660** may be configured to use Time Division Multiplex (TDM), Internet Protocol (IP), Ethernet, optical networking, wireless protocols, communication signaling, or some other communication format-including combinations thereof. Communication interface **660** may be configured to communicate with other computing systems (both physical and/or virtual) and a coordination service to obtain a configuration for computing system **600**.

[0038] Processing system **650** comprises microprocessor and other circuitry that retrieves and executes operating software from storage system **645**. Storage system **645** may include volatile and nonvolatile, removable, and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. Storage system **645** may be implemented as a single storage device but may also be implemented across multiple storage devices or sub-systems. Storage system **645** may comprise additional elements, such as a controller to read operating software from the storage systems. Examples of storage media include random access memory, read only memory, magnetic disks, optical disks, and flash memory, as well as any combination or variation thereof, or any other type of storage media. In some implementations, the storage media may be a non-transitory storage media. In some instances, at least a portion of the storage media may be transitory. In no case is the storage media a propagated signal.

[0039] Processing system **650** is typically mounted on a circuit board that may also hold the storage system. The operating software of storage system **645** comprises computer programs, firmware, or some other form of machine-readable program instructions. The operating software of storage system **645** comprises coordination communication service **630** and packet processing service **632**. The operating software on storage system **645** may further include an operating system, utilities, drivers, network interfaces, applications, or some other type of software. When read and executed by processing system **650**, the operating software on storage system **645** directs computing system **600** to operate as described herein.

[0040] In at least one implementation, coordination communication service **630** directs processing system **650** to communicate with a coordination service for a private network. The coordination service can be used to maintain and distribute communication configuration information that permits computing elements to communicate in the private network. The configuration information can include private networking information (private IP addresses, ports, and the like) for the computing elements, encryption key information (public encryption keys for the computing elements), public addressing information, or some other communication information. The configuration information can permit computing elements in the private network (or gateways supporting the private network) to encapsulate, decapsulate, and forward packets to other permitted devices in the network.

[0041] Packet processing service **632** directs processing system **650** to receive a first packet that encapsulates a second packet, wherein the second packet includes an IPv6 address. The IPv6 address can include a unique identifier for gateway computing system **600** relative to one or more other gateways supporting the private network, can include an IPv4 address for the computing element behind gateway computing system **600**, can include a flag indicating that packet translation is required, or can include some other information to support the processing of the second packet. When the first packet is received, packet processing service **632** directs processing system **650** to decapsulate the packet, wherein decapsulation may comprise applying a private encryption key to the second packet. Once decapsulated, packet processing service **632** can determine whether translation is required on the IPv6 address. In some examples, gateway computing system **600** can translate all incoming communications. In other examples, gateway computing system **600** can determine whether a flag is set in the IPv6 address to indicate a translation requirement. Once packet processing service **632** determines that translation is required, packet processing service **632** directs processing system **650** to identify the IPv4 address associated with the IPv6 address and replace the IPv6 address with the IPv4 address. The second packet is forwarded to a destination computing element, such as a virtual machine, with the IPv4 address in place of the IPv6 address. In some examples, the IPv6 address can include the IPv4 address for the destination.

[0042] In some implementations, in translating the IPv6 address to the IPv4 address, packet processing service **632** directs processing system **650** to translate the IPv6 header of the communication to an IPv4 header. The IPv4 header will include at least the identified destination IPv4 address and can further include a source IPv4 address.

[0043] FIG. **7** illustrates a coordination service computing system **700** to manage and distribute configuration information associated with a private network according to an implementation. Computing system **700** is representative of any computing system or systems with which the various operational architectures, processes, scenarios, and sequences disclosed herein for a coordination service can be implemented. Computing system **700** is an example of coordination service **120** of FIG. **1**, although other examples may exist. Computing system **700** includes storage system **745**, processing system **750**, and communication interface **760**. Processing system **750** is operatively linked to communication interface **760** and storage system **745**. Communication interface **760** may be communicatively linked to storage system **745** in some implementations. Computing system **700** may further include other components such as a battery and enclosure that are not shown for clarity.

[0044] Communication interface **760** comprises components that communicate over communication links, such as network cards, ports, radio frequency (RF), processing circuitry and software, or some other communication devices. Communication interface **760** may be configured to communicate over metallic, wireless, or optical links. Communication interface **760** may be configured to use Time Division Multiplex (TDM), Internet Protocol (IP), Ethernet, optical networking, wireless protocols, communication signaling, or some other communication format-including combinations thereof. Communication interface **760** is configured to communicate with physical and/or virtual computing elements in one or more private networks. The computing elements include gateways that can be used as an intermediary or relay for computing elements incapable of direct communication with other computing elements in the private network.

[0045] Processing system **750** comprises microprocessor and other circuitry that retrieves and executes operating software from storage system **745**. Storage system **745** may include volatile and nonvolatile, removable, and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. Storage system **745** may be implemented as a single storage device but may also be implemented across multiple storage devices or sub-systems. Storage system **745** may comprise additional elements, such as a controller to read operating software from the storage systems. Examples of storage media include random access memory, read only memory, magnetic disks, optical disks, and flash memory, as well as any combination or variation thereof, or any other type of storage media. In some implementations, the storage media may be a non-transitory storage media. In some instances, at least a portion of the storage media may be transitory. In no case is the storage media a propagated signal.

[0046] Processing system **750** is typically mounted on a circuit board that may also hold the storage system. The operating software of storage system **745** comprises computer programs, firmware, or some other form of machine-readable program instructions. The operating software of storage system **745** comprises coordination service **730** and relay service **735**. The operating software on storage system **745** may further include an operating system, utilities, drivers, network interfaces, applications, or some other type of software. When read and executed by processing system **750**, the operating software on storage system **745** directs computing system **700** to operate as described herein.

[0047] In at least one implementation, addressing and rule service **735** can maintain communication configuration information for a private network. The communication configuration information can include private addressing information associated with computing elements of the private network, public addressing information associated with computing elements of the private network, encryption keys, communication rules indicating permitted communications between the computing elements, or some other information. In some implementations, when a computing element is added to the network, the computing element can provide public addressing information, public encryption information, or some other information to the coordination service. The coordination service can associate a private IP address to each of the computing elements, wherein the private IP address uniquely identifies each of the computing elements.

[0048] In some implementations, a private network can employ gateways for VPCs or other datacenters, wherein the computing elements behind the gateway can use a subnet of IPv4 addresses. When multiple gateways use overlapping subnets, each of the gateways can be assigned a unique identifier that can separate packets directed at a first gateway from packets identified in a second gateway. Specifically, if a computing element, such as an administrator computer, initiated a request to communicate with a virtual machine executing behind a first gateway, the IPv6 private address for the communication will include the IPv4 address associated with the virtual machine, the unique identifier for the gateway, or some other information that directs the packet to the desired destination. When the packet is identified by the administrator computer, the packet can be encapsulated using a public key associated with the destination gateway and a public IP address can be included in the encapsulation header to forward the packet to the destination gateway. In this example, the computing element can use the gateway identifier to select the encryption key and destination address used for communicating the packet across the network.

[0049] Distribute service **730** directs processing system **700** to distribute configuration information to individual computing elements and gateways that provide the private network. When a new computing element joins the network, information about the new computing element can be distributed to other computing elements and gateways in the network. Additionally, communication information can be distributed to the newly joining computing element about the other computing elements previously existing in the private network. As an example, when a gateway is added that provides connectivity to a set of computing elements in an IPv4 subnet, coordination service may identify a unique identifier for the gateway. The unique identifier can be used to direct packets to the computing elements using the gateway. Information about the unique identifier, the subnet, encryption parameters, and the like can be distributed by distribute service **730** to other computing elements in the computing environment. Once distributed, a first computing element can direct a packet to a second computing element (e.g., virtual machine) behind the new gateway using an IPv6 address that incorporates at least the IPv4 address for the second computing element and the unique identifier for the gateway. The first computing element can also encapsulate the packet using the encryption parameters provided by the coordination service and the public addressing associated with the first computing element and the gateway. In at least one example, the computing elements and gateways can maintain one or more data structures to direct and encapsulate communications in the private network. In a VPC, the gateway can provide the encapsulation and forwarding operations for the virtual machines or containers supported by the gateway.

[0050] The included descriptions and figures depict specific implementations to teach those skilled in the art how to make and use the best mode. For teaching inventive principles, some conventional aspects have been simplified or omitted. Those skilled in the art will appreciate variations from these implementations that fall within the scope of the invention. Those skilled in the art will also appreciate that the features described above can be combined in various ways to form multiple implementations. As a result, the invention is not limited to the specific implementations described above, but only by the claims and their equivalents.

What is claimed is:

1. A method for sending a packet over a private network, the method comprising:

in an application executing on a source computing element:

obtaining an IPv6 destination address for a destination on the private network from a coordination service of the private network, wherein the IPv6 destination address includes an IPv4 destination address of the destination in an IPv4 subnet and a gateway identifier identifying a gateway to the IPv4 subnet;

generating the packet addressed to the IPv6 destination address; and

forwarding the packet to a private networking service on the source computing element; and

in the private networking service:

determining a public IP address for the gateway from the gateway identifier, wherein the coordination service indicates the public IP address corresponding to the gateway identifier;

encapsulating the packet to create an encapsulated packet addressed to the public IP address; and

sending the encapsulated packet to the gateway at the public IP address.

2. The method of claim **1**, wherein obtaining the IPv6 destination address comprises:

requesting the IPv6 destination address from a Domain Name System (DNS) for domains of the private network, wherein the DNS is controlled by the coordination service.

3. The method of claim **2**, wherein the DNS is implemented on the source computing element.

4. The method of claim **2**, wherein the DNS is implemented in another computing element of the private network.

5. The method of claim **1**, wherein the coordination service assigns the gateway identifier to the gateway and other identifiers to respective other gateways of the private network.

6. The method of claim **5**, wherein the IPv4 subnet overlaps with another subnet accessible via one of the other gateways.

7. The method of claim **1**, comprising:

receiving an encryption key for communications with the destination from the coordination service, wherein encapsulating the packet includes encrypting the packet using the encryption key.

8. The method of claim **1**, wherein the gateway decapsulates the encapsulated packet and transmits the packet to the IPv4 destination address on the IPv4 subnet.

9. The method of claim **8**, wherein the gateway extracts the IPv4 destination address from the IPv6 destination address upon determining address translation is necessary.

10. A apparatus for a source computing element to send a packet over a private network, the apparatus comprising:

a storage system;

a processing system operatively coupled to the storage system; and

program instructions stored on the storage system that, when executed by the processing system, direct the apparatus to:

receive IPv6 destination addresses for destinations on the private network from a coordination service for the private network, wherein each destination address includes an IPv4 address and a gateway identifier;

identify an IPv6 destination address of the IPv6 destination addresses for a destination of the destinations, wherein the IPv6 destination address includes an IPv4 destination address of the destination in an IPv4 subnet and a gateway identifier identifying a gateway to the IPv4 subnet;

generate a packet addressed to the IPv6 destination address;

determine a public IP address for the gateway from the gateway identifier using information provided by the coordination service;

encapsulate the packet to create an encapsulated packet addressed to the public IP address; and

send the encapsulated packet to the gateway at the public IP address.

11. The apparatus of claim **10**, wherein the program instructions direct the apparatus to:

provide a Domain Name System (DNS) on the apparatus, wherein the DNS receives the IPv6 destination addresses and the IPv6 destination address is identified from the DNS.

12. The apparatus of claim **10**, wherein the coordination service assigns the gateway identifier to the gateway and other identifiers to respective other gateways of the private network.

13. The apparatus of claim **12**, wherein the IPv4 subnet overlaps with another subnet accessible via one of the other gateways.

14. The apparatus of claim **10**, wherein the program instructions direct the apparatus to:

receive an encryption key for communications with the destination from the coordination service, wherein encapsulation of the packet includes encrypting the packet using the encryption key.

15. The apparatus of claim **10**, wherein the gateway decapsulates the encapsulated packet and transmits the packet to the IPv4 destination address on the IPv4 subnet.

16. The apparatus of claim **15**, wherein the gateway extracts the IPv4 destination address from the IPv6 destination address upon determining address translation is necessary.

17. One or more non-transitory computer readable storage media having program instructions stored thereon for sending a packet over a private network, the program instructions, when read and executed by a processing system, direct the processing system to:

receive IPv6 destination addresses for destinations on the private network from a coordination service for the private network, wherein each destination address includes an IPv4 address and a gateway identifier;

identify an IPv6 destination address of the IPv6 destination addresses for a destination of the destinations, wherein the IPv6 destination address includes an IPv4 destination address of the destination in an IPv4 subnet and a gateway identifier identifying a gateway to the IPv4 subnet;

generate a packet addressed to the IPv6 destination address;

determine a public IP address for the gateway from the gateway identifier using information provided by the coordination service;

encapsulate the packet to create an encapsulated packet addressed to the public IP address; and

send the encapsulated packet to the gateway at the public IP address.

18. The one or more non-transitory computer readable storage media of claim **17**, wherein the program instructions direct the processing system to:

provide a Domain Name System (DNS) on the processing system, wherein the DNS receives the IPv6 destination addresses and the IPv6 destination address is identified from the DNS.

19. The one or more non-transitory computer readable storage media of claim **17**, wherein the coordination service assigns the gateway identifier to the gateway and other identifiers to respective other gateways of the private network, and wherein the IPv4 subnet overlaps with another subnet accessible via one of the other gateways.

20. The one or more non-transitory computer readable storage media of claim **17**, wherein the program instructions direct the processing system to:

receive an encryption key for communications with the destination from the coordination service, wherein encapsulation of the packet includes encrypting the packet using the encryption key.

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